



The Psychology and Management of Project Teams: An Interdisciplinary Perspective

François Chiochio (ed.) et al.

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CHAPTER

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13 Functional Diversity in Project Teams: Working Across Boundaries

Sujin K. Horwitz

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Abstract

This chapter reviews literatures from organizational psychology and management to assess the impact of member diversity on project team performance. Specifically, among numerous diversity attributes discussed in the team literature, functional diversity and its impact on project team performance are closely examined in this chapter, as project teams typically consist of members from different functional expertise and backgrounds to accomplish complex, novel, and non-routine tasks. In doing so, two broad dimensions of functional diversity, inter- and intra-personal functional diversity, are delineated, and their effects on project team process and outcomes are discussed by reviewing extant team research from both organizational psychology and management. This chapter also examines the potential moderation of intrapersonal functional diversity on the relationship between overall functional diversity and project team performance, followed by directions for further research and implications for practice in project management.

Keywords: [project teams](#), [project team process](#), [project team performance](#), [functional diversity](#), [interpersonal functional diversity](#), [intrapersonal functional diversity](#)

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Introduction

As elucidated by Hobbs, Chiochio, and Kelloway in Chapter 1 of this volume, there has been a stream of multidisciplinary research to investigate various elements of project teams (Brown & Eisenhardt, 1995; Chiochio, Grenier, O'Neill, Savaria, & Willims, 2012). This chapter reviews literatures from management and organizational psychology to assess the impact of diversity on project team performance. Specifically, among numerous diversity attributes discussed in the team literature, functional diversity and its effects on project team performance are closely examined in this chapter, as project teams typically consist of members from different functional expertise and backgrounds to accomplish complex, novel, and non-routine tasks (Denison, Hart, & Kahn, 1996; Edmondson & Nembhard, 2009; McDonough, 2000). In doing so, two broad dimensions of functional diversity, inter- and intrapersonal functional diversity, are delineated, and their effects on project team process and outcomes are discussed by reviewing extant team research from both organizational psychology and management. This chapter also examines the potential moderation of intrapersonal functional diversity on the relationship between overall functional diversity and project team performance, followed by directions for further research and implications for practice.

p. 330 Classifications of Diversity in Extant Literature

Chiochio (Chapter 3 of this volume) defines the term “project teams” as the following:

A project team unites people with varied knowledge, expertise and experience who, within the life span of the project but over long work cycles, must acquire and pool vast amounts of information in order to define or clarify their purpose, adapt or create the means to progressively elaborate an incrementally or radically new concept, service, product, activity, or more generally, to generate change.

From the above definition, one significant element of project teams is the relevance of functional diversity, the extent to which team member attributes reflect knowledge, experience, and perspectives pertinent to accomplishing tasks. Additionally, it can be inferred that the impact of functional diversity on project team performance is likely to be affected by members' abilities to combine their unique expertise to contribute to project success. Note that numerous other diversity attributes can influence project teams in other ways than those comprising functional diversity examined in this chapter. For example, Harrison, Price, and Bell (1998) examined the impact of surface-level and deep-level diversity on team integration. The researchers defined surface-level diversity as member differences in visible biological characteristics (e.g., age, gender, and race/ethnicity) and deep-level diversity as differences in attitudes, beliefs, and values that are learned through member interactions. In their cross-sectional study of teamwork, Harrison, Price, Gavin, and Florey (2002) found that over time, the effects of surface-level diversity were less important, whereas the effects of deep-level diversity were more prominent in predicting group outcomes. Milliken and Martins (1996) similarly categorized diversity into two broad types, “observable individual differences” and “underlying attributes,” while Pelled (1996) dichotomized diversity based on “visibility” and “job-relatedness” dimensions. In Pelled's taxonomy of diversity, job relatedness was operationally defined as the extent to which the attribute reflects experience, skills, or perspectives pertinent to accomplishing tasks. In a similar vein, Joshi and Roh (2009) distinguished between relations-oriented (e.g., gender, ethnicity, and age) and task-oriented diversity (e.g., education, seniority, and functional expertise) to meta-analyze the effects of the two diversity categories on team performance. The researchers identified that functional diversity had a significant and positive effect on team performance as compared to other forms of task-oriented diversity. A meta-analysis by Horwitz and Horwitz (2007) also reported a positive impact of task-related diversity on team performance, whereas bio-demographic diversity was not significantly related to team performance. A more recent meta-analysis by Bell et al. (2011) focused on the effects of specific demographic variables (e.g., functional background, education

level, and organizational tenure) on team performance, rather than those of broad diversity categories (e.g., highly job-related, less visible), as typically studied in other meta-analyses. The researchers reported that a variety in functional backgrounds had a small yet positive relationship with general team performance as well as more specific dimensions of team performance, such as team creativity and innovation. Finally, a meta-analysis by Gerwin and Barrowman (2002) was particularly noteworthy, as they evaluated the relationships between project-salient integrative product development (IPD) characteristics and project performance. Gerwin and Barrowman identified that cross-functional teaming (i.e., creating a team consisting of functionally diverse members), one of IPD's main characteristics, significantly reduced a product development time in their meta-analysis.

Research endeavors to ascertain the effects of team diversity on teamwork have garnered prolific outcomes on the topic, and many studies have highlighted a positive relationship between task-related diversity and team performance, as discussed in the preceding section. Particularly, functional diversity has been suggested to benefit teams that perform novel and complex tasks across a variety of team contexts, as demonstrated in several meta-analyses (Devine & Philips, 2001; Bell et al., 2011; Gerwin & Barrowman, 2002). Because project teams are generally formed to undertake unique and complex tasks and their membership tends to be more functionally heterogeneous than that of other types of teams, functional diversity is assumed to hold the most significant impact on the complex dynamics of project team performance and thus is especially important to understand (Cohen & Bailey, 1997; Edmonson & Nembhard, 2009; Gemser & Leenders, 2011). However, the majority of team studies and especially meta-analyses tend to mix types of teams in their analyses; thus it is impossible to determine whether and to what extent the “generic” relationship seen for all types of teams is also applicable to project teams (Chiocchio & Essiembre, 2009). The ensuing section examines the multifaceted dimensions of functional diversity in the project team context.

Dimensions of Functional Diversity in Project Teams

Most of existing work on the diversity literature conceptualizes functional diversity as the distributional differences among individuals on a team with respect to varying functional areas and amounts of job-related expertise (Bunderson & Sutcliffe, 2002). Functional diversity has thus become generally operationalized as a multidimensional construct embracing a variety of differences in individual functional expertise, experience, and knowledge (Bunderson & Sutcliffe, 2002). Functionally diverse teams are often equated with teams consisting of members who possess in-depth knowledge in their respective functional areas; however, functional diversity can also be examined on an individual basis by measuring the breadth of functional experience that an individual possesses. This within-individual functional diversity is assessed by measuring whether an individual on a team is a narrow functional specialist with experience in a limited range of functions, or a broad generalist whose work experience spans a range of functional areas (Bunderson & Sutcliffe, 2002). Less work has investigated how the varying dimensions of functional diversity affect project team performance, despite the prolific research on teams in both organizational psychology and project management. The following section reviews the team literature to refine the current conceptualization of functional diversity.

Existing Frameworks Through a New Lens: Dichotomy of Functional Diversity

Clearly delineating functional diversity and its dimensions is critical to a discussion of how different dimensions of functional diversity affect project team performance. The current literature in organizational psychology reveals several existing frameworks to explain multiple facets of functional diversity. First, Bunderson and Sutcliffe (2002) classified functional diversity in four categories: (1) dominant function diversity, described as different functional areas within which team members have spent most of their careers; (2) functional background diversity, defined as differences in the complete functional backgrounds of team members; (3) functional assignment diversity, conceptualized as differences in functional assignments of team member; and (4) intrapersonal diversity, denoted as differences in functional backgrounds of individual team members (pp. 878–880). In contrast, Harrison and Klein (2007) offered three components of diversity in analyzing their effects on team outcomes: (1) separation, denoted as compositional differences among individuals in their lateral position with respect to values, beliefs, and attitudes; (2) variety, referred to as compositional differences among individuals with respect to some relevant categories that contribute to team diversity; and (3) disparity, conceptualized as vertical differences in proportion of valued assets or desirable resources, hence creating inequality or relative concentration within a team. Finally, Dawson (2011) expanded Harrison and Klein's typology by refining the construct of variety into two sub-dimensions: (1) range in variety, defined as the range of levels of a variable is represented within a group (p. 88); and (2) spread in variety, defined as the degree to which all possible levels of a variable are equally or evenly represented within a group (for further discussion of these concepts in relation to selecting project team members, see Allen & O'Neill, Chapter 12 of this volume).

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In this chapter, functional diversity is dichotomized into the following parsimonious dimensions: interpersonal functional diversity and intrapersonal functional diversity. Interpersonal functional diversity is defined as between-member differences in functional domains across team members, thereby including varying functional backgrounds, expertise, and knowledge at the group level. Intrapersonal functional diversity is conceptualized as within-person differences in functional domains in order to reflect the breadth of functional diversity that individual team members possess. Intrapersonal functional diversity, while frequently used as a group-level measure, also affects team performance, given the extent to which members' prior experiences are individually heterogeneous or homogeneous. Intrapersonal functional diversity thus embraces the notion of *spread* in functional varieties in teams by capturing both the range of functional experience within the individual, as well as the common representation of functional experiences among team members at the group level. For example, finding a high level of intrapersonal functional diversity in a team indicates that individual team members have broad functional experience that overlaps with each other's experience, thereby indicating a commonality in functional experience among the members at the group level. In contrast, a high level of interpersonal functional diversity in a team suggests that the team has a diverse knowledge base and expertise, with a large number of functional domains represented across team members. Although these two types of diversity fall under the same rubric of functional diversity, they tap into fundamentally different facets of functional bases, as interpersonal functional diversity pertains to the breadth of functional diversity at the group level, whereas intrapersonal functional diversity examines the breadth of functional diversity at the individual team member level. The concept of intrapersonal functional diversity may thus allow for a more specific analysis, hence elucidating the complex impact of functional diversity on project team dynamics. The next section reviews the extant literature on these two dimensions of functional diversity and assesses their effects on project team process and performance. For project team process, task conflict, external communication, and boundary spanning due to increased functional diversity are highlighted with a summary of relevant research findings, while the overall impact of functional diversity on project performance is examined with a particular emphasis on the effectiveness and efficiency of novel projects.

Literature Review on the Two Types of Functional Diversity in Project Teams

Relevant articles on the two categories of functional diversity in project teams published since 1990 were identified by computerized searches of academic databases and journals. The major keywords used for the searches were “project teams,” “new product development teams,” “project team diversity,” and “functional diversity/heterogeneity,” in order to narrow the vast amount of research done on teamwork in both organizational psychology and project management. Keyword combinations and truncation were also used to capture relevant studies on the effects of functional diversity on team outcomes in the project team context. Finally, the searches were limited to empirical studies on project teams performing real-life tasks in natural settings (i.e., real organizations), while excluding project teams performing simulated tasks in laboratory settings (i.e., educational institutions and training centers). A total of 22 empirical studies that examined the two functional diversity categories in the project team context were identified (see Table 13.1 for the summary of the identified studies). Of these studies, 18 studies examined the relationship between interpersonal functional diversity and project team outcomes, while four studies focused on the effects of intrapersonal functional diversity on project team performance. Additionally, four meta-analyses that assessed the cumulative effects of functional diversity on project performance were identified and included in this review.

Table 13.1 Effects of Interpersonal and Intrapersonal Functional Diversity on Project Team Performance

Authors	Journal	Sample	Main Results
Ancona & Caldwell (1992)	<i>Organization Science</i>	45 NPD teams	Positive effect of functional assignment diversity on external communication, which led to higher team and manager ratings of effectiveness
Eisenhardt & Tabrizi (1995)	<i>Administrative Science Quarterly</i>	72 multifunctional project teams	Interpersonal functional diversity was related to shorter product development time
Pelled et al. (1999)	<i>Administrative Science Quarterly</i>	45 NPD and process improvement teams	High levels of dominant functional diversity experienced greater task-related conflict but were not related to performance
Sethi (2000)	<i>Journal of Marketing</i>	141 cross-functional teams	No effect of functional diversity on the quality of NPD
Keller (2001)	<i>Academy of Management Journal</i>	93 R&D teams	Positive indirect effect of functional diversity on quality, efficiency, and budget, while negative indirect effect on cohesiveness through external communication
Lovelace et al. (2001)	<i>Academy of Management Journal</i>	43 NPD teams	No direct effect but significant moderating effects of interpersonal functional diversity on innovation
Sethi et al. (2001)	<i>Journal of Marketing Research</i>	141 cross-functional teams	No effect of functional diversity on the innovativeness of NPD
Cummings (2004)	<i>Management Science</i>	182 problem-solving teams	Functional assignment diversity was positively related to external knowledge sharing
Joshi & Sharma (2004)	<i>Journal of Marketing</i>	165 NPD projects	Creating cross-functional teams was positively related to customer knowledge development
Bstieler (2005)	<i>Journal of Product Innovation Management</i>	182 development projects	Significant effect of functional representation on time efficiency was found
Randel & Jaussi (2003)	<i>Academy of Management Journal</i>	37 cross-functional teams	Identification with other members on functional background is positively related to an individual member performance
Van Der Vegt & Bunderson (2005)	<i>Academy of Management Journal</i>	57 R&D teams	Significant moderating effects of functional assignment diversity on team learning and performance

Yeh & Chou (2005)	<i>Social Behavior and Personality</i>	88 cross-functional teams	Support for main effects but no support for the moderating effect of functional background diversity
Carbonell & Rodriguez (2006)	<i>Journal of Business Research</i>	83 projects	Functional diversity positively influenced innovation speed and the effect was stronger for complex projects than simple projects
Liang et al. (2007)	<i>Industrial Management & Data Systems</i>	16 project teams	Functional diversity indirectly increased team performance through increased task conflict
Akgun et al. (2008)	<i>Innovation & Management</i>	207 NPD projects	Functional diversity was positively related to a team's ability to manage and integrate knowledge
Dayan & Di Benedetto (2009)	<i>European Journal of Innovation Management</i>	93 NPD teams	Functional diversity was quadratically (a $\cap$ shape) related to team communication, coordination, and cohesion
Haon et al. (2009)	<i>Marketing Letters</i>	142 NPD teams	Functional diversity had a positive effect on NPD performance mediated by the degree of instrumental use of information
Park et al. (2009)	<i>Journal of Product Innovation Management</i>	62 NPD teams	Multi-knowledge as intrapersonal functional diversity positively influenced NPD performance
Huckman & Staats (2011)	<i>Production and Operations Management</i>	409 software development teams	Intrapersonal diversity positively affected team performance when a coordination of a team task was required due to change

Interpersonal Functional Diversity in Project-Team Processes

Interpersonal functional diversity, the extent to which team members differ in their dominant functions, backgrounds, and assignments, was by far the most studied dimension of functional diversity in project teams. A review of interpersonal functional diversity in project teams reveals two critical processes in facilitating project team performance: a high level of interpersonal functional diversity is associated with increased task conflict and expanding communication and boundaries in project teams.

Effects of Interpersonal Functional Diversity on Task Conflict

The current literature posits that a moderate level of task-based conflict can produce positive effects by generating ideas, evaluating alternatives, and encouraging constructive criticism, all of which can improve the quality of team performance (De Dreu & Weingart, 2003; Jehn, 1995; Klein & Harrison, 2007). In particular, when tasks are of a non-routine nature and are complex, as in project

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teams, conflict regarding team task can enhance the depth of problem evaluation and assessment of options (Amason & Mooney, 1999; Jehn & Mannix, 2001). This finding was consistent with several studies examining new product development (NPD) and innovation in project teams, demonstrating that task conflict positively affects the relationship between functional diversity and project team outcomes. When task conflict is properly managed, task conflict stimulates creativity and innovation through encouraging divergent viewpoints and thus represents an advantage that functionally diverse teams have relative to homogenous teams (Jehn & Bendersky, 2003; Liang et al., 2007; Pelled et al., 1999). Findings from research suggest that task conflict is related to team performance in a curvilinear fashion in that performance is highest when task conflict is moderate, while extreme levels of task conflict are associated with a decrease in performance (De Dreu, 2007; Van de Vliert & De Dreu, 1994) (for more on conflict in the project context, see de Wit, Chapter 9 of this volume).

Pelled, Eisenhardt, and Xin (1999) studied the impact of team diversity on task conflict and performance by surveying 45 new product and process improvement teams in electronic divisions at three companies. Although dominant functional diversity was not significantly related to team performance in their sample, Pelled et al. (1999) reported that teams with high dominant functional diversity experienced greater task-related conflict, with the relationship being moderated by team longevity and task routineness. Their findings suggest that task conflict has a positive relationship with cognitive task performance because such conflict fosters a deeper understanding of task issues and exchange of information, which ultimately facilitates problem-solving and idea generation. There is also a moderating effect of team longevity on task conflict in that when team tenure is less, the relationship between dominant functional diversity and task conflict tends to be weakened (Pelled et al., 1999). Similarly, Liang et al. (2007) investigated the effects of functional assignment background (i.e., team member diversity in departmental and unit affiliations in an organization) among 16 software project teams and reported that such diversity indirectly increased team performance through increased task conflict. Finally, by examining 43 cross-functional NPD teams in a global high tech industry, Lovelace et al. (2001) found that functional assignment diversity in teams positively influenced the effectiveness and efficiency of innovation through the mediating effect of team communication. Specifically, the researchers noted that a greater level of functional diversity increases levels of task disagreement, but the effect of task disagreement on teams' outcomes depends on how such disagreement is communicated, how free team members feel to express task-related issues, and the ability of the teams to attain creative solutions to their problems in the process of resolving such conflict.

Effects of Interpersonal Functional Diversity on Crossing Boundaries and Expanding Communication

The current project management research postulates that functional diversity in projects teams enhances members' ability to connect and communicate with multiple sources that are internal and external to teams, thereby integrating relevant resources to benefit team performance (Argote, Ingram, Levine, & Moreland, 2000; Edmondson & Nembhard, 2009; Marrone, Tesluk, & Carson, 2007). In particular, external communication allows a project team to import relevant information into the team and coordinate internal activities as needed (Edmondson, 1999). The team literature from both organizational psychology and project management highlights that heterogeneous team members in terms of functional backgrounds, affiliations, and organizational positions are efficient at crossing boundaries and expanding networks, as such functionally diverse members can act as boundary spanners to facilitate information transfer and intersect inter- and intra-organizational boundaries (Cummings, 2004; Richter, West, Van Dick, & Dawson, 2006). Such effective boundary spanning and networking activities are crucial for project teams, as their tasks are largely interdependent of the rest of the organization, thereby necessitating the coordination of resources and cooperation from multiple stakeholders.

Surveying 182 teams engaged in problem-solving projects, Cummings (2004) reported that an interaction of external knowledge sharing and functional assignment diversity was positively related to team performance in that greater external information sharing was more strongly associated with team performance when teams were more functionally diverse. Observing 45 product development teams, Ancona and Caldwell (1992) also found that successful product development teams took part in a variety of boundary-spanning and political activities, including coordinating tasks, seeking support and feedback, and promoting the team's image. Communication strategy was also found to be germane to team performance in that the most successful product development teams used a variety of external communication strategies and task-coordination behaviors, which helped these teams obtain task-related information, secure resources, and enhance project success. In contrast, less successful product development teams used fewer types of external communication activities and prolonged information-seeking. This study demonstrated the importance of task-oriented external communication and coordination behaviors, hence suggesting that product development teams benefit not only from greater frequency of external communication, but also from the quality of such interactions (Ancona & Caldwell, 1992). Keller (2001) likewise argued that cross-functional team composition offers a unique advantage to teams by providing multiple sources of communication and network, both inside and outside an organization. In his study of 93 R & D teams at four companies, Keller identified that functional assignment diversity had a positive and indirect effect on team outcomes through external communication. That is, although functional diversity itself had an insignificant (even a negative effect) on team outcomes, cross-functional teams delivered better technical quality, faster schedule performance, and significantly more cost savings when mediated through enhanced external communication with stakeholders. Keller's study stressed that cross-functional teams can enhance project outcomes through increased external communication because the benefits are due to having members with diverse backgrounds and skills, coupled with contacts with important external networks of information. Similarly, Akgun et al. (2008) found that the number of functional areas represented in a project team is positively related to the team's ability to acquire, process, and utilize knowledge from various sources, and that doing so increases the team's flexibility and responsiveness to changes and customer demands.

Impact of Interpersonal Functional Diversity on Project Performance

In the current literature, project performance criteria include a variety of performance measures (Atkinson, 1999; Shenhar, Tishler, Dvir, Lipovetsky, & Lechler, 2002; Turner, 2009). Although schedule, cost, and quality, commonly referred to the “iron triangle,” have been frequently used to measure project team performance (Olsen, 1971), divergent viewpoints exist to embrace varying dimensions of project team performance given the multidisciplinary nature of projects. Among various performance dimensions discussed in the project management literature, effectiveness and efficiency of novel projects are examined in this chapter, as functionally diverse project teams are generally well equipped to develop innovative solutions when problems and tasks are complex, ill-defined, and novel (Mohrman, Cohen, & Mohrman, 1995; Simons, Pelled, & Smith, 1999; West, 2002).

Interpersonal Functional Diversity and Effectiveness/Efficiency of Novel Projects

Numerous researchers have argued that cognitive diversity, as reflected in the breadth and depth of functional diversity, provides a particularly rich source for creativity and innovation in teams (McDonough, 2000; Milliken & Martins, 1996; Pelled, Eisenhardt, & Xin, 1999; Zhou & Shalley, 2010). For example, proponents of the cognitive resource hypothesis argue that cognitive diversity enhances a team’s collective ability to solve complex problems and achieve novel projects (Hambrick, Cho, & Chen, 1996; Miller, Burke, & Glick 1998). Similarly, the value-in-diversity paradigm stresses that diversity in organizations brings net-added value to organizational processes and outcomes (Cox & Blake, 1991). As reflected in these perspectives, researchers maintain that teams composed of members with varying skills, knowledge, and expertise will be more innovative than teams composed of homogeneous members because a variety of functionality in teams generates unique yet complementary perspectives and experience on issues and problems (Miller, Burke, & Glick, 1998). Indeed, a large body of research suggests that functional characteristics positively influence product development and innovation in project teams (Edmonson & Nembhard, 2009; Gerwin & Barrowman, 2002). Additionally, some further contend that if relevant departments (e.g., engineering, marketing, and production) are involved from the beginning of a project, the activities in these departments can concurrently be coordinated to allow an efficient transfer of information to facilitate project coordination (Griffin, 2002).

Several studies have focused on the effects of interpersonal functional diversity on NPD and innovation process in the cross-cultural context by examining project teams sampled from different countries. For example, Bstieler (2005) studied team characteristics and contextual factors affecting project timeliness with 182 development projects in Australia and Canada and demonstrated that one of the key factors for project timeliness was the number of functions represented in a project team, coupled with the members’ dedication and integration into the project. Joshi and Sharma (2004) surveyed 165 Canadian marketing managers as key informants for cross-functional NPD projects to test antecedents, moderators, and outcomes of customer knowledge development process. The researchers uncovered that cross-functional NPD teams fostered an understanding of customer preference, which was essential to the success of NPD, hence underscoring the importance of functional diversity in the NPD process. Finally, a study by Eisenhardt and Tabrizi (1995) examined 72 multifunctional product teams in 36 computer firms located in Japan, Europe, and the United States and discovered that cross-functional teams created a wider range of ideas and expedited the product development process because downstream problems in the various functional areas were more likely to be identified and corrected during early stages in the production process due to the involvement of multiple units. Another noteworthy finding of the study was that an experiential and improvisational strategy through frequent iterations, increased testing, establishing milestones, and utilizing powerful leadership expedited the product development process, particularly in uncertain and complex product innovation environments. Their findings mirror the current thought of project management, agile project delivery, which employs

incremental delivery and iterative process of working products for project modification and enhancement, as discussed by Hobbs (Chapter 2 of this volume). Although a life cycle of a given project should be adapted to characteristics and contexts unique to each project, an agile approach seems to be well-suited for experimental, innovative, and novel projects in a dynamic, changing environment because the approach incorporates incremental changes and progressive elaborations throughout the product development cycle (Eisenhardt & Tabrizi, 1995).

Haon, Gotteland, and Fornerino (2009) embraced the notion of competence diversity, a more inclusive measure of functional diversity, by adding education, experience, and expertise to functional diversity and examining its effects on NPD performance. Haon and colleagues reported that competence diversity had a positive effect on NPD performance mediated by the degree of instrumental use of information. Carbonell and Rodriguez (2006) operationalized functional diversity as the number of functional areas and external stakeholders represented on a team (e.g., top management, marketing, engineering, manufacturing, sales, etc.) and measured its effect on the speed of product development in various industries. The authors classified the sample into two categories based on the technical complexity of a project (simple or complex), and performed a separate analysis for each sub-sample. Their results indicated that functional diversity had a curvilinear relationship with innovation speed for both technologically complex and simple projects. As functional diversity increased, so did the speed of product development; however, when functional diversity became too high, there was then a reduction in innovation speed. Additionally, there was a varying degree of acceleration in innovation speed between the two types of project in that the more complex and challenging the project was, the greater functional interdependence was needed to expedite its execution, whereas functional diversity was less beneficial for simple projects requiring clear solutions (Carbonell & Rodriguez, 2006). Similarly, Van Der Vegt and Bunderson (2005) investigated how forming collective identification moderated the effects of functional assignment diversity on team learning and outcomes (for more on identity in project teams, see Tremblay et al., Chapter 8 of this volume). The researchers identified that at the extreme ends of the continuum, either too little or too much, of functional assignment diversity among team members inhibited team learning and decreased team performance. Van Der Vegt and Bunderson highlighted that moderate levels of functional assignment diversity in teams made them more conducive to team learning and performance. Finally, Dayan and Di Benedetto (2009) surveyed 93 NPD team managers in Turkey to examine the effects of functional diversity, along with several other antecedents on the quality of team interactions (teamwork quality). The researchers reported that ↪ high teamwork quality was achieved when functional diversity increased from a low to a moderate level. As suggested by the aforementioned studies, the impact of member diversity on team performance is likely to be affected by the complexity and structural aspects of the task (Van de Ven & Ferry, 1980; Dumont, Gibson Jr., & Fish, 1997). For example, in accomplishing a highly complex and ill-defined task, it is necessary for team members to pull together their diverse expertise to formulate strategies to deal with the task under complex conditions, whereas member diversity may be unnecessary or even counterproductive in dealing with a simple, routine task (Horwitz, 2005). There should be an optimal match between team diversity and the scope and complexity of a project to ensure project team success (Dumont, Gibson Jr., & Fish, 1997).

Empirical studies have also reported either negative or null effects of interpersonal functional diversity on project team performance. For example, Sethi, Smith, and Park (2001) surveyed 141 cross-functional team managers regarding how new product quality was affected by project team characteristics, such as functional background diversity and information integration among team members. From their results, Sethi et al. proposed that diverse functional inputs would facilitate the development of a new product and further hypothesized that the quality of a new product would be highest when functional background diversity in a cross-functional team is moderate. Contrary to their expectations, functional background diversity in itself had no effect on the quality of a new product. Yeh and Chou (2005) examined the effects of functional diversity on the performance of 88 cross-functional teams implementing a sophisticated, system-wide software program at a high-tech firm. The researchers expected that functional diversity would be positively associated with team performance, while team conflict would mediate the effect of functional diversity on performance. However,

like Sethi et al.'s study, neither a main effect of functional diversity nor a mediating effect of task conflict was found for their cross-functional team sample with respect to having a significant impact on team performance. In other words, a high degree of functional diversity was found to be neither the main source of task conflict nor a contributing factor for project effectiveness. There were also several meta-analyses summarizing the impact of functional diversity specifically on project team outcomes, and their findings were similarly inconsistent regarding the benefits of functional diversity. For example, a meta-analytic review by Henard and Szymanski (2001) found that NPD was not significantly correlated with cross-functional integration, thereby suggesting that an integration of multiple functional areas into the new product initiative may not necessarily improve the success of new products. Likewise, Chen, Damanpour, and Reilly (2010) meta-analyzed the relationships between NPD speed and 17 antecedents of NPD, including functional diversity, and did not establish a significant relationship between functional diversity and NPD speed. More recently, Sivasubramaniam et al. (2012) reported an insignificant relationship between functional diversity and NPD success in terms of efficiency (marketing budgets and schedules), effectiveness (market success), and speed to market in their meta-analysis. Another noteworthy study on the impact of functional diversity on project teamwork was conducted by Zoogah, Vora, Richard, and Peng (2011) by surveying 44 strategic alliance teams drawn from both US and non-US companies. Contrary to their expectations, Zoogah et al. found negative effects of functional assignment diversity on both performance satisfaction and goal achievement and speculated that this result may have been caused by the lack of distinction between interpersonal and intrapersonal functional diversity in their analysis. The research team reasoned that intrapersonal functional diversity as reflected in the functional specialists in team composition may have engendered a negative effect on team outcomes in their sample.

Summary of Empirical Studies on the Overall Impact of Interpersonal Functional Diversity on Project Teams

The underlying logic of the benefits of interpersonal functional diversity in project teams is twofold. First, from the intra-group process perspective, interpersonal functional diversity has a positive impact on project team performance because of unique yet complementary functional expertise and experience that members bring to the team, which leads to synergetic team outcomes (Hambrick, Cho, & Chen, 1996; Horwitz & Horwitz, 2007). Second, from the intergroup process perspective, having diverse team members regarding functional backgrounds, assignments, and affiliations in organizations enables a project team to span boundaries and gain access to key resources from both inside and outside the team in order to pool and assimilate such resources to augment their project (Ancona & Caldwell, 1992; Caldwell & O'Reilly, 1982). However, the totality of empirical research on the benefits of interpersonal functional diversity for project performance has presented a complex picture and inconsistent findings. On one hand, from both theoretical perspectives (notably from the cognitive resource and value-in-diversity perspectives) and empirical bases, by broadening the range of human capital among team members, such varieties in interpersonal functional diversity can promote the effectiveness of project team performance. Indeed, this review has found a number of empirical studies supporting the perspective that functionally heterogeneous project teams are more innovative in developing products and respond more quickly to market demands than homogenous ones. However, not all studies concur, and the results are inconclusive. In some cases, negative effects have actually been found between functional diversity and project team performance, as discussed in the preceding section.

Although the impact of functional diversity is paradoxical to some extent within the project team context, some of the equivocal findings on the effects of functional diversity may be due to the fact that extreme levels of such diversity produce negative consequences, whereas moderate levels lead to optimal performance, as consistent with the curvilinear effects found in the team literature. That is, it is possible that project outcomes may be compromised when each member is far more knowledgeable in one area relative to other members, as shown for high levels of interpersonal functional diversity suggested by several studies (Bunderson & Sutcliffe,

2002; Zoogah et al., 2011). Conversely, having team members with broad functional experience that overlaps with each other (high intrapersonal functional diversity) may be instrumental in stimulating the exchange and integration of information and reducing interpersonal conflict, as the members are more likely to have a shared understanding and relate to each others' experience due to their broad exposure to various functions in the organization (Bunderson, 2003; Byul, Boone, Hendriks, & Matthyssens, 2011; Maznevski, 1994). This chapter proposes that project team outcomes can be more accurately understood by including intrapersonal functional diversity as a potential moderator in the relationship between interpersonal functional diversity and project team outcomes.

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As shown in Figure 13.1, task conflict, external communication, boundary spanning, and information sharing (project team process) are theorized as team mechanisms through which interpersonal functional diversity facilitates distal outcomes of project teams. The effects of interpersonal functional diversity on project team outcomes are thus mediated by the project team process, as depicted in the figure. Additionally, the mediated relationship between interpersonal functional diversity and project team outcomes varies across levels of intrapersonal functional diversity, as suggested in the moderated mediation in the figure. This proposition, therefore, predicts that project team process mediates the relationship between interpersonal functional diversity and project performance at different levels of the moderator, intrapersonal functional diversity. On the basis of this proposition, it is thus argued that a high level of intrapersonal functional diversity improves team process as compared with a low level of intrapersonal functional diversity, and the improved team process positively mediates the relationship between interpersonal functional diversity and project team outcomes. Because intrapersonal functional diversity represents another crucial dimension of functional diversity in assessing its impact on project team dynamics and has been largely ignored in the current literature, a more comprehensive assessment of the benefits and limitations of functional diversity in project teams is needed, whereby intrapersonal functional diversity is tested as a moderator in understanding the complex relationship between functional diversity and project team outcomes.

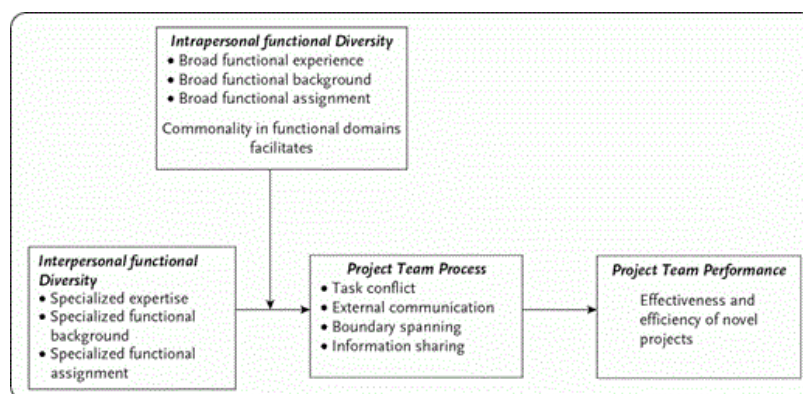


Figure 13.1 The moderation of intrapersonal functional diversity in the mediated relationship between interpersonal functional diversity and project team performance

Intrapersonal Functional Diversity and Its Moderating Impact on Project-Team Dynamics

The majority of empirical studies on project team diversity have used a one-dimensional approach in conceptualizing functional diversity, as opposed to taking a multidimensional approach that accounts for both interpersonal and intrapersonal functional differences (Bunderson & Sutcliffe, 2002). By failing to isolate the potential effects of intrapersonal functional diversity from the broader operationalization of functional diversity, the lesser influence of such results can be directly tied to project team performance. Additionally, there is an inherent assumption that individuals are capable of integrating others' information with little misunderstanding, which seems erroneous, given the potential for relationship conflict ensuing from such misunderstandings (Cronin & Weingart, 2007; Jehn, Chatwick, & Thatcher, 1997; Jehn, Northcraft, & Neale, 1999). A tendency also exists to overlook the political nature of project team cooperation, which can hinder the development of team cohesion. As illustrated in the preceding chapters, project teams often involve cross-functional members with different values, interests, and objectives (Hoever, Knippenberg, Ginkel, & Barkema, 2012; Nabukenya, Van Bommel, Proper, & De Vreede, 2011). For example, in developing and launching a new product in an organization, individual members on a NPD team can have conflicting objectives due to their departmental affiliation. It is important for a sales team member to push for an aggressive product launch date and competitive pricing to boost sales, whereas a production unit member is more concerned with estimating prudent costs so that production can meet scheduling and quality goals. Such various perspectives and conflicting interests can create problems and can hinder cross-functional collaboration in project teams (Ancona & Caldwell, 1992). Furthermore, negative effects of functional diversity on project team collaboration are more likely the result of parochialistic views and member emphasis on hasty categorizations of individual differences rather than project-relevant information (Byrne, Clore, & Worchel, 1966; Tajfel, & Turner, 1986; Tziner, 1985; Williams & O'Reilly, 1998). A team's ability to integrate project-relevant information and knowledge is largely influenced by both commonalities and differences in the individually held occupation and contextual knowledge (Dougherty, 1992). Individuals trained in a particular discipline, profession, or occupation have substantial conceptual and practical knowledge in common with others from the same discipline or occupation, while sharing common language, mental models, and practices (Cronin & Weingart, 2007; Parker, Atkins, & Axtell, 2008). Conversely, unfamiliar language, different mental schema, and idiosyncratic practices of heterogeneous team members can lead to difficulty in communication, synthesis, and agreement on individual members' informational contributions to benefit project team outcomes (Dougherty & Hardy, 1996; Parker & Axtell, 2001; Wiersema & Bantel, 1992).

Finally, to aggravate the negative effects of categorizations and parochialism, there is an additional factor affecting project team performance: the temporal dimensions of project teams in terms of membership, duration, and process, as discussed by Chiochio (Chapter 3 of this volume). Project teams generally have relatively limited project duration, coupled with fluid membership, as they are assembled for performing specific tasks within a finite time frame signaled by explicit deadlines and time constraints. Such temporariness reduces potential opportunities for team members to gain member familiarity and shared understanding, which is more prevalent in long-standing work teams (Edmonson & Nembhard, 2009; Haon et al., 2009). In addition, project teams tend to lack sufficient team temporal duration, the length and frequency of member interactions, which has been found to mitigate negative effects of team diversity on collaboration (Harrison et al., 2002; Mohammed, Hamilton, & Lim, 2009). Finally, member attrition and turnover at varying temporal points impact project teams in a substantially negative manner. As illustrated by Chiochio (Chapter 3 of this volume), although turnover across project phases is problematic regardless of project team types, the damaging effects of unforeseen turnover are much more pronounced in integrated *do-it-all* teams than component project teams since such member attrition impacts the complex role that integrated *do-it-all* project teams assume in managing and executing projects during multiple work cycles. Therefore, capitalizing on diverse functional resources in project teams is critically dependent upon each member's ability to

understand, communicate, and integrate the ideas of other team members to accomplish a unified task in a relatively short project timeline. Research has consistently found that multidisciplinary, cross-functional knowledge integration requires higher levels of team members' involvement in information sharing; however, the more functionally diverse the team members are, the less common knowledge and shared understanding among their functional differences are likely to be (Caldwell & O'Reilly, 2003; Mesmer-Magnus & DeChurch, 2012). This review, therefore, flowing from the extensive yet somewhat conflicting literature, has found that considering these obstacles that potentially inhibit effective team functioning, teams consisting of functional generalists (i.e., members with broad functional backgrounds) will be able to facilitate team process and performance more effectively than teams largely composed of functional specialists with narrowly defined functional expertise (Kessler & Chakrabarti, 1999; van Knippenberg, De Dreu, & Homan, 2004). The ensuing section elucidates the potential benefits of intrapersonal functional diversity on project teams by highlighting key empirical findings from both organizational psychology and project management.

Effects of Intrapersonal Functional Diversity on Project Teams

An empirical study by Park, Lim, and Birnbaum-More (2009) described intrapersonal functional diversity by using the concept called "multi-knowledge," in which members understand multiple functional areas as distinguished from mono-knowledge, a depth of expertise in one's own functional domain in the absence of knowledge of others' functional areas in a team. Using the tenets of expectancy theory (Vroom, 1964), Park et al. (2009) asserted that teams largely composed of multi-knowledge individuals are more likely to understand the capabilities and expertise of other team members, particularly in team-based projects. In such cases, individuals have greater ability to share knowledge with others, which in aggregate should lead to an overall high level of information sharing and understanding within the team. Using 62 cross-functional NPD teams drawn from multiple firms in South Korea, the researchers confirmed that the proportion of multi-knowledge individuals on cross-functional NPD teams had a positive effect on product innovativeness through the information-sharing process, thereby confirming the benefits of intrapersonal functional diversity in project team performance. Similarly, Richter et al. (2012) proposed shared understanding of others' expertise, "knowledge of who knows what (KWKW)" in capturing the socially shared "meta-knowledge" (p. 1283) of individual members' expertise within teams, which is similar to the notion of transactive memory in team mental models (Austin, 2003; Wegner, 1986). The researchers suggest that a team's awareness of an individual's unique knowledge and its source, as reflected in KWKW, can benefit team performance through combining the knowledge possessed by each team member. Surveying 34 multidisciplinary pharmaceutical R&D teams, Richter and colleagues investigated how functionally diverse teams could facilitate their member creativity by serving as an information reservoir. The researchers found that as team members had more divergent expertise and knowledge, shared KWKW became more crucial in improving an individual member's creativity by facilitating the individual's use of diverse expertise and knowledge of other members. Conversely, for teams with low functional diversity, where unique and dispersed expertise within the team was limited, shared KWKW did not benefit individuals in their creative endeavors as strongly as teams with high functional diversity. Extending on this line of research, Randel and Jaussi (2003) investigated 37 cross-functional teams and discovered that an individual's perceived identity related to functional background, deemed "functional social identity," explained more variance of an individual's performance as a cross-functional team member than relational demography that simply considers an actual degree of dissimilarity in functional background as a factor to explain such variance. Specifically, the individual's identification with other members from the same functional background is positively related to his or her self-esteem and subsequent performance as a cross-functional team member, and the individual is thus more likely to engage in cooperative behaviors to enhance the attractiveness of that functional background.

Finally, in a study of 409 software development projects at a software services firm in India, Huckman and Staats (2011) dichotomized diversity into two broad categories with respect to team members' customer

experience: intrapersonal experience diversity, defined as individual team members' breadth of experience in interacting with customers, and interpersonal experience diversity, conceptualized as team members' depth of experience in interacting with customers in completing projects. The researchers reported that a team's intrapersonal diversity in customer experience positively affected team performance, particularly when a coordination of a team task was required to change task requirements over the course of a project. Conversely, increasing the interpersonal difference in customer experience across team members negatively impacted team performance under conditions of task change and subsequent coordination. Based on their findings, Huckman and Staats (2011) contend that intrapersonal diversity is more beneficial for project team performance in volatile and uncertain project environments than stable environments, as such team composition of members with a breadth of functional experience enhances a project team's ability to adapt and respond efficiently to changes that require the coordination and integration demands on team members.

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Summary of Empirical Studies on the Effects of Intrapersonal Functional Diversity on Project Team Outcomes

Despite the scarcity of studies examining the effects of intrapersonal functional diversity in the project team context, the preceding review nonetheless suggests that intrapersonal functional diversity provides an important common ground for team members to communicate, coordinate, and integrate their unique and diverse expertise in project teams, in which wide functional differences can acutely heighten interpersonal differences between team members. The study by Huckman and Staats (2011) is noteworthy in this regard, as they found that teams composed of individuals with a vastly different experience (i.e., high interpersonal diversity) were challenged with coordination issues and thus delayed team processes to complete projects. Alternatively, teams with high intrapersonal diversity delivered projects more efficiently when tasks changed rapidly, thereby supporting that intrapersonal diversity facilitates efficacious coordination and integration of tasks for functionally heterogeneous teams. Although interpersonal aspects of functional diversity can have negative implications for team functioning, this review has observed the opposite for intrapersonal functional diversity by pointing to a significant and positive impact of such diversity on functionally diverse project teams through facilitating team processes. Additionally, it should be noted that the construct of intrapersonal diversity has been increasingly addressed in studies of project teams, although the construct has been operationalized somewhat differently, such as multi-knowledge (Park et al., 2009), intrapersonal experience diversity (Huckman & Staats, 2011), and KWKW (Richter et al., 2012). In summary, the current research seems to favor positive effects of intrapersonal functional diversity on project team performance by suggesting that team members with a broad experience and understanding of the relationships among different functions on a project are uniquely positioned to utilize member expertise to improve team processes for synergistic outcomes.

Directions for Future Research

There has been a growing interest in investigating various factors affecting the efficacy of project teams in multiple fields; functional diversity is one of such factors studied in an interdisciplinary fashion, as demonstrated by the extensive scope of research in both organizational science and project management. As reflected in the subject areas of the studies included in this review, research on project teams is inherently a multidisciplinary endeavor, encompassing production and operations management to organizational psychology and marketing (see Table 13.1 for the list of journals). Therefore, research opportunities to advance the current understanding of project team diversity are multidisciplinary and abundant. In particular, this review observed three key areas that future research should address.

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First, as a significantly sparse amount of research on the effects of intrapersonal diversity on project teams has been conducted and the ramifications are important, this is an area in need of further examinations. Although the construct of intrapersonal diversity has been increasingly addressed in team research, the current literature still severely lacks empirical studies investigating the moderating role of intrapersonal functional diversity on project team interactions and outcomes. For example, several articles examining this important issue were identified; however, the majority of them are either conceptual pieces on intrapersonal functional diversity or empirical studies examining the effect of top management teams' intrapersonal diversity on firm performance, which differs considerably from project teams discussed in this chapter (Bunderson 2003; Cannella, Park, & Lee, 2008; Davis, Bell, Payne, & Kreiser, 2010). Therefore, the significance of intrapersonal functional diversity for project team outcomes is largely understudied in the current literature, hence calling for greater stream of research on this area.

A second important line of research lies in further exploration of how the duration of project team membership can change the dynamics of functional diversity. A number of studies postulated non-monotonic, curvilinear models to explain the effects of functional diversity on team outcomes when moderated by team longevity. Team temporal duration, for example, is a mechanism to help teams develop member familiarity, as team members have the opportunity to engage in meaningful interactions over time (Harrison, Price, & Bell, 1998; Mohammed, Hamilton, Lim, 2009). However, there remains a paucity of studies investigating the temporal relationship between intrapersonal functional diversity and project team outcomes. Intrapersonal functional diversity in the context of project team longevity is a particularly crucial area to be examined because project teams' fluid membership, coupled with relatively short team tenure, implies that member familiarity and understanding derived from team longevity are generally absent. In the absence of team longevity, intrapersonal functional diversity may help team members recognize and integrate their diverse knowledge for efficient task completion. Only a handful of studies investigated the moderating role of intrapersonal functional diversity on team processes, and none looked at the effects of intrapersonal functional diversity on team dynamics with a longitudinal lens. There is thus an important need for research examining the dynamic moderation played by intrapersonal functional diversity in temporary and time-bound project team performance.

Finally, more research is needed to identify and test other potential moderators affecting the relationship between team diversity and project outcomes. Although this review largely focused on the moderating role of intrapersonal functional diversity, other important moderating variables exist in project team diversity. In particular, several researchers called for more examination on contextual moderators and other intervening variables, such as environmental uncertainty (low vs. high), organization types (manufacturing vs. service), and innovation types (incremental vs. radical innovation), that potentially affect the efficacy of project work in functionally diverse teams (Chen, Damanpour, & Reilly, 2010; Miles, 2005). Another fruitful research area is to examine the moderating impact of task interdependence on the relationship between functional diversity and project performance, as functional diversity may not impact teams with low levels of task interdependence (e.g., pooled or sequential task interdependence) as strongly as teams working on highly interdependent and reciprocally integrated projects (Thompson, 1967). The current literature, however, lacks studies investigating the moderating role of task interdependence in the functional diversity–project team performance relationship (Chiocchio & Essiembre, 2009). At the same time, it is evident that there is ample room for more studies on the effects of functional diversity in project team settings. In spite of the prolific research done on the topic, when studies were categorized under each dimension of functional diversity, there were only a few examining intrapersonal functional diversity.

It should be acknowledged that this chapter was limited to the examination of functional diversity and its effects on the effectiveness and efficiency of novel projects. There are, however, other diversity attributes that impact various dimensions of project performance, as noted previously. While the focus on functional diversity in project teams allowed for a more comprehensive assessment of the complex relationship between such

diversity and project performance, it also limits considerations of the impact and implications of other important diversity dimensions in the project team context. Therefore, future research should seek to uncover other specific elements of team diversity and their effects on project teams.

Implications for Project Management

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Unlike innate and immutable bio-demographic diversity at the individual level, functional diversity is more malleable at both the individual and team levels in team composition. For example, employees can transfer from one functional unit to another to broaden their functional experience, and in practice, an increasing number of companies utilize cross-training to obtain organizational flexibility, broaden employee skills, and thus increase the value of their employees' human capital (Marks, Sabella, Burke, & Zacaro, 2002; Volpe, Cannon-Bowers, Salas, & Spector, 1996). Similar to cross-training, another managerial strategy that counteracts the potential negative impact of functional diversity on project team collaboration is cross-cut role assignments, in which individuals are simultaneously members of more than one task group or team for a fixed duration of time (Bettencourt & Dorr, 1998). For example, Anthro, an Oregon-based furniture manufacturer, has been successfully utilizing a cross-cut role assignment program called "Shadow Program," to improve internal relations and cooperation among teams and departments. In Anthro's Shadow Program, employees shadow other employees to understand different functional roles and develop interdepartmental relationships to improve their teamwork (Layne, 2000). As the primary goal of cross-training and cross-cut role assignments is to enhance knowledge of interpersonal activities by introducing team members to the roles and responsibilities of their teammates, such training can improve team interaction, communication, and coordination among diverse team members and thus help organizations capitalize on the cross-functionality of project teams (Blickensderfer, Cannon-Bower, & Salas, 1998).

Additionally, organizations can build shared knowledge by making information on individual skills and expertise easily available and accessible to employees. For example, an organization can promote knowledge sharing by cataloging information on employee expertise and making such information available across the organization (Spreitzer, 2006). An employee can then access the database of employee expertise to search and consult others with relevant expertise when needed. Indeed, with the expansion of information technology, organizations are increasingly utilizing HR information databases to identify, store, and retrieve knowledge, skills, and abilities of employees. Such employee skills databases greatly enhance the organization's capability to track and integrate employee talent for creating project teams with optimal complementary skills.

Finally, caution should be exercised when promoting member diversity in project teams. Simply increasing the amount of functional diversity in teams is not sufficient to improve project team performance. Instead, the characteristics of the specific project must be taken explicitly into account to maximize the benefits of functional diversity. In doing so, a right combination of interpersonal and intrapersonal functional diversity makes it more likely that members will utilize their different perspectives with their teams to optimize their project performance.

Final Thought

This chapter challenges project team researchers as well as practitioners to rethink how functional diversity can impact project team performance by refining the existing concept into interpersonal and intrapersonal functional diversity and contrasting key findings of the empirical research on the two dimensions. The refinement of the construct in particular was captured by the notion of intrapersonal functional diversity, defined as variation in functional domains within individual members, which is different from the traditional construct of functional diversity discussed in the project team literature. Drawing on team research from organizational psychology and project management, this work also proposed a conceptual model in which intrapersonal functional diversity moderates the relationship between functional diversity and project performance and thus presented both opportunities and challenges of managing functional diversity in project teams. The inclusion of intrapersonal functional diversity as a moderator in the functional diversity–team process–team outcomes relationship allows for a more consistent analysis and accurate identification of the impact of functional diversity on project team outcomes.

For those who support that functionally diverse members benefit project performance, a myopic, single-lens approach to understanding the complex nature of such diversity may miss the mark, as this review suggests that the effects of functional diversity are not uniformly positive, while engendering some contradictory findings. It is no doubt that as functional diversity increases in a project team, so does the breadth of knowledge, expertise, and experience that the team can collectively utilize to enhance the quality and quantity of projects. However, it has been also found that functional diversity, combined with complex tasks and temporal structures in project teams, reduces member familiarity and shared understanding, leading to the potential for communication difficulties, coordination challenges, and suboptimal knowledge integration. This review contends that teams with high levels of intrapersonal functional diversity have a broader repertoire of experience to draw upon, share a larger proportion of their functional background, and thus have a better understanding of different functions and how they are related to accomplish team goals.

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References

- Amason, A. C., & Mooney, A. C. (1999). The effects of past performance on top management team conflict in strategic decision making. *International Journal of Conflict Management*, 10, 340–359.
[Google Scholar](#) [WorldCat](#)
- Ancona, D. G., & Caldwell, D. F. (1992). Demography and design: Predictors of new product team performance. *Organizational Science*, 3(3), 321–342.
[Google Scholar](#) [WorldCat](#)
- Akgun, A., Dayan, M., & Di Benedetto, A. (2008). Antecedents and consequences of team intelligence. *Information & Management*, 45(4), 221–226.
[Google Scholar](#) [WorldCat](#)
- Argote, L., Ingram, P., Levine, J. M., & Moreland, R. L. (2000). Knowledge transfer in organizations: Learning from the experience of others. *Organizational Behavior and Human Decision Processes*, 82(1), 1–8.
[Google Scholar](#) [WorldCat](#)
- Atkinson, R. (1999). Project management: Cost, time and quality, two best guesses and a phenomenon, it's time to accept other success criteria. *International Journal of Project Management*, 17(6), 337–342.
[Google Scholar](#) [WorldCat](#)
- Austin, J. R. (2003). Transactive memory in organizational groups: The effects of content, consensus, specialization, and accuracy on group performance. *Journal of Applied Psychology*, 88(5), 866–878.
[Google Scholar](#) [WorldCat](#)
- Bell, S. T., Villado, A. J., Lukasik, M. A., Belau, L., & Briggs, A. L. (2011). Getting specific about demographic diversity variable and team performance relationships: A meta-analysis. *Journal of Management*, 37(3), 709–743.
[Google Scholar](#) [WorldCat](#)
- Bettencourt, B. A., & Dorr, N. (1998). Cooperative interaction and intergroup bias: Effects of numerical representation and cross-cut role assignment. *Personality and Social Psychology Bulletin*, 24, 1270–1287.
[Google Scholar](#) [WorldCat](#)
- Blickensderfer, E., Cannon-Bowers, J. A., & Salas, E. (1998). Cross-training and team performance. In J. A. Cannon-Bowers & E. Salas (Eds.), *Making decisions under stress: Implications for individual and team training* (pp. 299–311). Washington, DC: American Psychological Association.
[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)
- Brown, S. A., & Eisenhardt, K. M. (1995). Product development: Past research, present findings, and future research. *Academy of Management Review*, 20(2), 343–378.
[Google Scholar](#) [WorldCat](#)
- Bstieler L. (2005). The moderating effect of environmental uncertainty on new product development and time efficiency. *Journal of Product Innovation Management*, 22, 267–284.
[Google Scholar](#) [WorldCat](#)
- Bunderson, J. S. (2003). Team member functional background and involvement in management teams: Direct effects and the moderating role of power centralization. *Academy of Management Journal*, 46(4), 458–474.
[Google Scholar](#) [WorldCat](#)
- Bunderson, J. S., & Sutcliffe, K. M. (2002). Comparing alternative conceptualizations of functional diversity in management teams: Process and performance effects. *Academy of Management Journal*. 45(5), 875–893.
[Google Scholar](#) [WorldCat](#)

Byrne, D., Clore, G., & Worchel, P. (1966). The effect of economic similarity-dissimilarity as determinants of attraction. *Journal of Personality and Social Psychology*, 4(2), 220–224. [↵](#)

[Google Scholar](#) [WorldCat](#)

Byul, T., Boone, C., Hendriks, W., & Matthyssens, P. (2011). Top management team functional diversity and firm performance: The moderating role of CEO characteristics. *Journal of Management Studies*, 48(1), 151–177.

[Google Scholar](#) [WorldCat](#)

Caldwell, D. F., & O'Reilly, C. A. (1982). Boundary spanning and individual performance: The impact of self-monitoring. *Journal of Applied Psychology*, 67(1), 124–127.

[Google Scholar](#) [WorldCat](#)

Caldwell, D. F., & O'Reilly, C. A. (2003). The determinants of team-based innovation in organizations: The role of social influence. *Small Group Research*, 34(4), 497–517.

[Google Scholar](#) [WorldCat](#)

Cannella, A. A., Park, J., & Lee, L. (2008). Top management team diversity and firm performance: Examining the roles of external and internal context. *Academy of Management Journal*, 51(4), 768–784.

[Google Scholar](#) [WorldCat](#)

Carbonell, P., & Rodriguez, A. I. (2006). Designing teams for speedy product development: The moderating effect of technological complexity. *Journal of Business Research*, 59(2), 225–232.

[Google Scholar](#) [WorldCat](#)

Chen, J., Damanpour, F., & Reilly, R. R. (2010). Understanding antecedents of new product development speed: A meta-analysis. *Journal of Operations Management*, 28, 17–33.

[Google Scholar](#) [WorldCat](#)

Chiocchio, F., & Essiembre, H. (2009). Cohesion and performance: A meta-analytic review of disparities between project teams, production teams, and service teams. *Small Group Research*, 40(4), 382–420.

[Google Scholar](#) [WorldCat](#)

Chiocchio, F., Grenier, S., O'Neill, T. A., Savaria, K., & Willms, D. J. (2012). The effects of collaboration on performance: A multilevel validation in project teams. *International Journal of Project Organisation and Management*, 4(1), 1–37.

[Google Scholar](#) [WorldCat](#)

Cohen, S. G., & Bailey, D. E. (1997). What makes teams work: Group effectiveness research from the shop floor to the executive suite. *Journal of Management*, 23(3), 230–290.

[Google Scholar](#) [WorldCat](#)

Cox, T., & Blake, S. (1991). Managing cultural diversity: Implications for organizational competitiveness. *Academy of Management Executive*, 5(3), 45–56.

[Google Scholar](#) [WorldCat](#)

Cronin, M. A., & Weingart, L. R. (2007). Representational gaps, information processing, and conflict in functionally diverse teams. *Academy of Management Review*, 32(3), 761–774.

[Google Scholar](#) [WorldCat](#)

Cummings, J. N. (2004). Work groups, structural diversity, and knowledge sharing in a global organization. *Management Science*, 50(3), 352–364.

[Google Scholar](#) [WorldCat](#)

Davis, J. L., Bell, R. G., Payne, G. T., & Kreiser, P. M. (2010). Entrepreneurial orientation and firm performance: The moderating role of managerial power. *The American Journal of Business*, 25(2), 41–54.

[Google Scholar](#) [WorldCat](#)

Dawson, J. F. (2011). *Measurement of work group diversity*. Unpublished doctoral dissertation. Aston University, Birmingham, England.

p. 357 Dayan, M., & Di Benedetto, C. A. (2009). Antecedents and consequences of teamwork quality in new product development teams. *European Journal of Innovation Management*, 12(1), 129–155. ↵
[Google Scholar](#) [WorldCat](#)

De Dreu, C. K. W. (2007). Cooperative outcome interdependence, task reflexivity and team effectiveness: A motivated information processing approach. *Journal of Applied Psychology*, 92(3), 628–638.
[Google Scholar](#) [WorldCat](#)

De Dreu, C. K. W., & Weingart, L. R. (2003). Task versus relationship conflict, team performance, and team member satisfaction: A meta-analysis. *Journal of Applied Psychology*, 88(4), 741–749.
[Google Scholar](#) [WorldCat](#)

Denison, D. R., Hart, S. L., & Kahn, J. A. (1996). From chimney to cross-functional teams: Developing and validating a diagnostic model. *Academy of Management Journal*, 39(4), 1005–1024.
[Google Scholar](#) [WorldCat](#)

Devine, D. J., & Philips, J. L. (2001). Do smart teams do better: A meta-analysis of cognitive ability and team performance. *Small Group Research*, 32(5), 507–532.
[Google Scholar](#) [WorldCat](#)

Dougherty, D. (1992). Interpretive barriers to successful product innovations in large firms. *Organization Science*, 3(2), 179–202.
[Google Scholar](#) [WorldCat](#)

Dougherty, D., & Hardy, C. (1996). Sustained product innovation in large, mature organizations: Overcoming innovation-to-organization problems. *Academy of Management Journal*, 39(5), 1120–1153.
[Google Scholar](#) [WorldCat](#)

Dumont, P., Gibson, G., Jr., & Fish, J. (1997). Scope management using project definition rating index. *Journal of Management in Engineering*, 13(5), 54–60.
[Google Scholar](#) [WorldCat](#)

Edmondson, A. C. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly*, 44(2), 350–383.
[Google Scholar](#) [WorldCat](#)

Edmondson, A. C., & Nembhard, I. M. (2009). Product development and learning in project teams: The challenges are the benefits. *Journal of Product Innovation Management*, 26(2), 123–138.
[Google Scholar](#) [WorldCat](#)

Eisenhardt, K. M., & Tabrizi, B. N. (1995). Accelerating adaptive processes: Product innovation in the global computer industry. *Administrative Science Quarterly*, 4(1), 84–110.
[Google Scholar](#) [WorldCat](#)

Gemser, G., & Leenders, M. A. A. M. (2011). Managing cross-functional cooperation for new project development success. *Long Range Planning*, 44, 26–41.
[Google Scholar](#) [WorldCat](#)

Gerwin, D., & Barrowman, N. J. (2002). An evaluation of research on integrated product development. *Management Science*, 48(7), 938–953.
[Google Scholar](#) [WorldCat](#)

Griffin A. (2002). Product development cycle time for business to business products. *Industrial Marketing Management*, 31(4),

291–304.

[Google Scholar](#) [WorldCat](#)

Hambrick, D. C., Cho, T. S., & Chen, M. J. (1996). The influence of top management team heterogeneity on firms' competitive moves. *Administrative Science Quarterly*, 41(4), 659–684.

[Google Scholar](#) [WorldCat](#)

Haon, C., Gotteland, D., & Fornerino, M. (2009). Familiarity and competence diversity in new product development teams: Effects on new product performance. *Marketing Letters*, 20(1), 75–89.

[Google Scholar](#) [WorldCat](#)

p. 358 Harrison, D. A., & Klein, K. J. (2007) What's the difference? Diversity constructs as separation, variety, or disparity in organizations. *Academy of Management Review*, 32(4), 1199–1228. ⁴

[Google Scholar](#) [WorldCat](#)

Harrison, D. A., Price, K. H., & Bell, M. P. (1998). Beyond relational demography: Time and the effects of surface- and deep-level diversity on work group cohesion. *Academy of Management Journal*, 41(1), 96–107.

[Google Scholar](#) [WorldCat](#)

Harrison, D. A., Price, K. H., Gavin, J. H., & Florey, A. T. (2002). Time, teams, and task performance: Changing effects of surface- and deep-level diversity on group functioning. *Academy of Management Journal*, 45(5), 1029–1045.

[Google Scholar](#) [WorldCat](#)

Henard, D. M., & Szymanski, D. M. (2001). Why some new products are more successful than others. *Journal of Marketing Research*, 38(3), 362–375.

[Google Scholar](#) [WorldCat](#)

Hoeber, I. J., Knippenberg, D. V., Ginkel, W. P. V., & Barkema, H. G. (2012). Fostering team creativity: Perspective taking as key to unlocking diversity's potential. *Journal of Applied Psychology*, 97(5), 982–996.

[Google Scholar](#) [WorldCat](#)

Horwitz, S. K. (2005). The compositional impact of team diversity on performance: Theoretical consideration. *Human Resource Development Review*, 4(2), 219–245.

[Google Scholar](#) [WorldCat](#)

Horwitz, S. K., & Horwitz, I. B. (2007). The effects of team diversity on team outcomes: A meta-analytic review of team demography. *Journal of Management*, 33(6), 987–1015.

[Google Scholar](#) [WorldCat](#)

Huckman, R. S., & Staats, B. R. (2011). Fluid tasks and fluid teams: The impact of diversity in experience and team familiarity. *Manufacturing & Service Operations Management*, 13(3), 310–328.

[Google Scholar](#) [WorldCat](#)

Jehn, K. A. (1995). A multimethod examination of the benefits and detriments of intragroup conflict. *Administrative Science Quarterly*, 40(2), 256–282.

[Google Scholar](#) [WorldCat](#)

Jehn, K. A., & Bendersky, C. (2003). Intragroup conflict in organizations: A contingency perspective on the conflict-outcome relationship. In R. M. Kramer & B. M. Staw (Eds.), *Research in organizational behavior* (Vol. 25, pp. 187–242). New York: Elsevier.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Jehn, K. A., Chatwick, C., & Thatcher, S. M. B. (1997). To agree or not to agree: The effects of value congruence, individual demographic dissimilarity, and conflict on workgroup outcomes. *International Journal of Conflict Management*, 8(4), 287–305.

[Google Scholar](#) [WorldCat](#)

Jehn, K. A., & Mannix, E. A. (2001). The dynamic nature of conflict: A longitudinal study of intragroup conflict and group performance. *Academy of Management Journal*, 44(2), 238–251.

[Google Scholar](#) [WorldCat](#)

Jehn, K. A., Northcraft, G. B., & Neale, M. A. (1999). Why differences make a difference: A field study of diversity, conflict, and performance in workgroups. *Administrative Science Quarterly*, 44(4), 741–763.

[Google Scholar](#) [WorldCat](#)

Joshi, A., & Roh, H. (2009). The role of context in work team diversity research: A meta-analytic review. *Academy of Management*, 44(3), 547–555.

[Google Scholar](#) [WorldCat](#)

p. 359 Joshi, A. W., & Sharma, S. (2004). Customer knowledge development: Antecedents and impact on new product performance. *Journal of Marketing*, 68(4), 47–59. ⁴

[Google Scholar](#) [WorldCat](#)

Keller, R. T. (2001). Cross-functional project groups in research and new product development: Diversity, communications, job stress, and outcomes. *Academy of Management*, 52(3), 599–627.

[Google Scholar](#) [WorldCat](#)

Kessler, E. H., & Chakrabarti, A. K. (1999). Speeding up the pace of new product development. *Journal of Product Innovation Management*, 16, 231–247.

[Google Scholar](#) [WorldCat](#)

Klein, K. J., & Harrison, D. A. (2007). On the diversity of diversity: Tidy logic, messier realities. *Academic of Management Perspectives*, 21(4), 26–33.

[Google Scholar](#) [WorldCat](#)

Layne, A. (2000, August 31). Walk a mile in my shadow. *Fast Company*, 38. Retrieved from <http://www.fastcompany.com/63150/walk-mile-my-shadow> (accessed October 12, 2012).

[WorldCat](#)

Liang, T., Liu, C., Lin, T., & Lin, B. (2007). Effect of team diversity on software project performance. *Industrial Management & Data Systems*, 107(5), 636–653.

[Google Scholar](#) [WorldCat](#)

Lovelace, K., Shapiro, D. L., Weingart, L. R. (2001). Maximizing cross-functional new product teams' innovativeness and constraint adherence: A conflict communications perspective. *Academy of Management Journal*, 44(4), 779–793.

[Google Scholar](#) [WorldCat](#)

Marks, M. A., Sabella, M. J., Burke, C. S., & Zaccaro, S. J. (2002). The impact of cross-training on team effectiveness. *Journal of Applied Psychology*, 87(1), 3–13.

[Google Scholar](#) [WorldCat](#)

Marrone, J. A., Tesluk, P. E., & Carson, J. B., (2007). A multilevel investigation of the antecedents and consequences to team member boundary spanning. *Academy of Management Journal*, 50(6), 1423–1439.

[Google Scholar](#) [WorldCat](#)

Maznevski, M. L. (1994). Understanding our differences: Performance in decision-making groups with diverse members. *Human Relations*, 47(5), 531–552.

[Google Scholar](#) [WorldCat](#)

McDonough, E. F. (2000). Investigation of factors contributing to the success of cross-functional teams. *Journal of Product Innovation Management*, 17(3), 221–235.

[Google Scholar](#) [WorldCat](#)

Mesmer-Magnus, J. R., & DeChurch, L. A. (2012). The cognitive underpinnings of effective teamwork: A meta-analysis. *Journal of Applied Psychology*, 95(1), 32–53.

[Google Scholar](#) [WorldCat](#)

Miles, I. (2005). Innovation in services. In J. Fargerbergm, D. C. Mowery, & R. R. Nelson (Eds.), *The Oxford handbook of innovation* (pp. 433–458). Oxford: Oxford University Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Miller, C. C., Burke, L. M., & Glick, W. H. (1998). Cognitive diversity among upper-echelon executives: Implications for strategic decision processes. *Strategic Management Journal*, 19(1), 39–58.

[Google Scholar](#) [WorldCat](#)

Milliken, F. J., & Martins, L. L. (1996). Searching for common threads: Understanding the multiple effects of diversity in organizational groups. *Academy of Management Review*, 21(2), 402–433.

[Google Scholar](#) [WorldCat](#)

Mohammed, S., Hamilton, K., & Lim, A. (2009). The incorporation of time in team research: Past, current, and future. In E. Salas, G. F. Goodwin, & C. S. Burke (Eds.), *Team effectiveness in complex organizations: Cross-disciplinary perspectives and approaches* (pp. 321–348). New York: Routledge Taylor and Francis Group. ↵

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Mohrman, S. A., Cohen, S., & Mohrman, A., Jr. (1995). *Designing team-based organizations*. San Francisco, CA: Jossey-Bass.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Nabukenya, J., Van Bommel, P., Proper, H. A., & De Vreede, G. J. (2011). An evaluation instrument for collaborative processes: Application to organizational policy-making. *Group Decision and Negotiation*, 20(4), 465–488.

[Google Scholar](#) [WorldCat](#)

Olsen, R. P. (1971). Can project management be designed? *Project Management Quarterly*, 2(1), 12–14.

[Google Scholar](#) [WorldCat](#)

Parker, S. K., & Axtell, C. M. (2001). Seeing another view point: Antecedents and outcomes of employee perspective-taking. *Academy of Management Journal*, 44(6), 1085–1100.

[Google Scholar](#) [WorldCat](#)

Parker, S. K., Atkins, P. W. B., & Axtell, C. M. (2008). Building better work places through individual perspective taking: A fresh look at a fundamental human process. In G. Hodgkinson & K. Ford (Eds.), *International review of industrial and organizational psychology* (Vol. 23, pp. 149–196). Chichester, UK: Wiley.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Park, M. H., Lim, J. W., & Birnbaum-More, P. (2009). The effect of multi-knowledge individuals on performance in cross-functional new product development teams. *The Journal of Product Innovation Management*, 26, 86–96.

[Google Scholar](#) [WorldCat](#)

Pelled, L. H. (1996). Demographic diversity, conflict, and work group outcomes: An intervening process theory. *Organizational Science*, 7(6), 615–631.

[Google Scholar](#) [WorldCat](#)

Pelled, L. H., Eisenhardt, K. M., & Xin, K. R. (1999). Exploring the black box: An analysis of work group diversity, conflict, and performance. *Administrative Science Quarterly*, 44(1), 1–28.

[Google Scholar](#) [WorldCat](#)

Randel, A. E., & Jaussi, K. S. (2003). Functional background identity, diversity, and individual performance in cross-functional teams. *Academy of Management Journal*, 46(6), 763–774.

[Google Scholar](#) [WorldCat](#)

Richter, A. W., Hirst, G., van Knippenberg, D., & Baer, M. (2012). Creative self-efficacy and individual creativity in team contexts: Cross-level interactions with team informational resources. *Journal of Applied Psychology*, 97(6), 1282–1290.

[Google Scholar](#) [WorldCat](#)

Richter, A. W., West, M. A., Van Dick, R., & Dawson, J. F. (2006). Boundary spanners' identification, intergroup contact and effective intergroup relations. *Academy of Management Journal*, 49(6), 1252–1269.

[Google Scholar](#) [WorldCat](#)

Sethi, R. (2000). New product quality and product development teams. *Journal of Marketing*, 64(2), 1–14.

[Google Scholar](#) [WorldCat](#)

Sethi, R., Smith, D., & Park, W. (2001). Cross-functional product development teams, creativity, and the innovativeness of new consumer products. *Journal of Marketing Research*, 23, 73–85.

[Google Scholar](#) [WorldCat](#)

Shenhar, A. J., Tishler, A., Dvir, D., Lipovetsky, S., & Lechler, T. (2002). Refining the search for project success factors: A multivariate, typological approach. *R&D Management*, 32(2), 111–126,

[Google Scholar](#) [WorldCat](#)

p. 361 Simons, T., Pelled, L. H., & Smith, K. A. (1999). Making use of difference: Diversity, debate, and decision comprehensiveness in top management teams. *Academy of Management Journal*, 42(6), 662–674. ⁴

[Google Scholar](#) [WorldCat](#)

Sivasubramaniam, N., Liebowitz, J. S., & Lackman, C. L. (2012). Determinants of new product development team performance: A meta-analytic review. *Journal of Product Innovation management*, 29(5), 803–820.

[Google Scholar](#) [WorldCat](#)

Spreitzer, G. M. (2006). Leading to grow and growing to lead: Leadership development lessons from positive organizational studies. *Organizational Dynamics*, 232, 1–12.

[Google Scholar](#) [WorldCat](#)

Tajfel, H., & Turner, J. C. (1986). The social identity theory of inter-group behavior. In S. Worchel & L. W. Austin (Eds.), *Psychology of intergroup relations* (pp. 7–24). Chicago: Nelson-Hall

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Thompson, J. (1967). *Organizations in action: Social science bases of administrative theory*. New York: McGraw-Hill.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Turner, R. (2009). *The handbook of project-based management* (3rd ed.). New York: McGraw-Hill.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Tziner, A. (1985). How team composition affects task performance: Some theoretical insights, *Psychological Reports*, 57, 1111–1119.

Van De Vliert, E., & De Dreu, C. K. W. (1994). Optimizing performance by stimulating conflict. *International Journal of Conflict Management*, 5(3), 211–222.

[Google Scholar](#) [WorldCat](#)

Van de Ven, A. H., & Ferry, D. L. (1980). *Measuring and assessing organizations*. New York: Wiley.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Van Der Vegt, G. S., & Bunderson, J. S. (2005). Learning and performance in multidisciplinary teams: The importance of collective team identification. *Academy of Management Journal*, 48(3), 532–547.

[Google Scholar](#) [WorldCat](#)

Van Knippenberg, D., De Dreu, C. K. W., & Homan, A. C. (2004). Work group diversity and group performance: An integrative model and research agenda. *Journal of Applied Psychology*, 89(6), 1008–1022.

[Google Scholar](#) [WorldCat](#)

Volpe, C. E., Cannon-Bowers, J. A., Salas, E., & Spector, P. E. (1996). The impact of cross-training on team functioning: An empirical investigation. *Human Factors*, 38, 87–100.

[Google Scholar](#) [WorldCat](#)

Vroom, V. H. (1964). *Work and motivation*. San Francisco, CA: Jossey-Bass.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Wegner, D. M. (1986). Transactive memory: A contemporary analysis of the group mind. In B. Mullen & G. R. Goethals (Eds.), *Theories of group behavior* (pp. 185–208). New York: Springer-Verlag.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

West, M. A. (2002). Sparkling foundations or stagnant ponds: An integrative model of creativity and innovation implementation in work groups. *Applied Psychology*, 51(3), 355–387.

[Google Scholar](#) [WorldCat](#)

Wiersema, M. F., & Bantel, K. A. (1992). Top management team demography and corporate strategic change. *Academy of Management Journal*, 35(1), 91–121.

[Google Scholar](#) [WorldCat](#)

Williams, K. Y., & O'Reilly, C. A. (1998). Demography and diversity in organizations. In B. M. Staw & R. M. Sutton. (Eds.), *Research in organizational behavior* (pp. 77–140). Stanford, CT: JAI Press.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

p. 362 Yeh, Y., & Chou, H. (2005). Team composition and learning behaviors in cross-functional teams. *Social Behavior and Personality*, 33(4), 391–402. ↵

[Google Scholar](#) [WorldCat](#)

Zhou, J., & Shalley, C. E. (2010). Deepening our understanding of creativity in the workplace: A review of different approaches to creativity research. In S. Zedeck (Ed.), *APA handbook of industrial and organizational psychology* (Vol. 1, pp. 275–302).

Washington, DC: American Psychological Association.

[Google Scholar](#) [Google Preview](#) [WorldCat](#) [COPAC](#)

Zoogah, D., Vora, D., Richard, O., & Peng, M. (2011). Strategic alliance team diversity, coordination, and effectiveness.

International Journal of Human Resource Management, 22(3), 510–529.

[Google Scholar](#) [WorldCat](#)